

```
MySQL 8.0 Command Line Client
-> P.name AS passenger_name,
-> P.social_network,
-> P.social_network_id,
-> Tr.train_name,
-> Tr.departure,
-> Tr.destination,
-> T.seat_number,
-> T.status
-> FROM Tickets T
-> JOIN Passengers P ON T.passenger_id = P.passenger_id
-> JOIN Trains Tr ON T.train_id = Tr.train_id;
+-----+-----+-----+-----+-----+-----+-----+-----+
| ticket_id | passenger_name | social_network | social_network_id | train_name | departure | destination | seat_number | status |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | John Doe | Facebook | john.fb123 | Express A | New York | Washington | 12A | Booked |
| 2 | Jane Smith | Twitter | jane_tw | Local B | Boston | New York | 5B | Booked |
+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT
-> P.name, P.social_network_id, T.ticket_id, Tr.train_name, T.seat_number
-> FROM Tickets T
-> JOIN Passengers P ON T.passenger_id = P.passenger_id
-> JOIN Trains Tr ON T.train_id = Tr.train_id
-> WHERE P.social_network = 'Facebook';
+-----+-----+-----+-----+-----+
| name | social_network_id | ticket_id | train_name | seat_number |
+-----+-----+-----+-----+-----+
| John Doe | john.fb123 | 1 | Express A | 12A |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> UPDATE Tickets
-> SET status = 'Cancelled'
-> WHERE ticket_id = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> DELETE FROM Tickets
-> WHERE ticket_id = 2;
```

```
MySQL 8.0 Command Line Client
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE AddStudent (
->     IN name VARCHAR(100),
->     IN course_name VARCHAR(50),
->     IN score INT
-> )
-> BEGIN
->     INSERT INTO Students (student_name, course, marks)
->     VALUES (name, course_name, score);
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL AddStudent('Alice', 'Computer Science', 85);
Query OK, 1 row affected (0.01 sec)

mysql> CALL AddStudent('Bob', 'Mechanical Engineering', 78);
Query OK, 1 row affected (0.01 sec)

mysql> CALL AddStudent('Charlie', 'Civil Engineering', 92);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Students;
```

student_id	student_name	course	marks
1	Alice	Computer Science	85
2	Bob	Mechanical Engineering	78
3	Charlie	Civil Engineering	92

Windows taskbar at the bottom shows the Start button, a search bar, and various application icons including File Explorer, Microsoft Edge, and the Task View button. The system clock in the bottom right corner displays 10:56 AM on 9/2/2025.

```
MySQL 8.0 Command Line Client

mysql> CREATE TABLE Students (
->   student_id INT AUTO_INCREMENT PRIMARY KEY,
->   student_name VARCHAR(100),
->   admission_date DATE
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE StudentAudit (
->   audit_id INT AUTO_INCREMENT PRIMARY KEY,
->   student_id INT,
->   action VARCHAR(50),
->   action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_student_insert
-> AFTER INSERT ON Students
-> FOR EACH ROW
-> BEGIN
->   INSERT INTO StudentAudit (student_id, action)
->   VALUES (NEW.student_id, 'INSERT');
-> END; //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Students (student_name, admission_date) VALUES ('Alice', '2025-09-01');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM StudentAudit;
+-----+-----+-----+-----+
| audit_id | student_id | action | action_time |
+-----+-----+-----+-----+
| 1 | 1 | INSERT | 2025-09-02 09:54:35 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```



```
mysql> select database();
+-----+
| database() |
+-----+
| strings    |
+-----+
1 row in set (0.00 sec)

mysql> SELECT CONCAT('Bus', SPACE(5), 'Ticket') AS result;
+-----+
| result          |
+-----+
| Bus      Ticket |
+-----+
1 row in set (0.00 sec)

mysql> SELECT SUBSTR('HyderabadToBangalore', 1, 9) AS result;
+-----+
| result          |
+-----+
| Hyderabad      |
+-----+
1 row in set (0.00 sec)

mysql> SELECT UPPER('bus ticket booking') AS result;
+-----+
| result          |
+-----+
| BUS TICKET BOOKING |
+-----+
1 row in set (0.01 sec)

mysql> SELECT LOWER('BUS TICKET BOOKING') AS result;
+-----+
| result          |
+-----+
| bus ticket booking |
+-----+
1 row in set (0.01 sec)

mysql> SELECT TRIM('   BusTicket   ') AS result;
+-----+
| result          |
+-----+
| BusTicket       |
+-----+
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line Client
-> RETURNS CHAR(1)
-> DETERMINISTIC
-> BEGIN
->   DECLARE grade CHAR(1);
->
->   IF score >= 80 THEN
->     SET grade = 'A';
->   ELSEIF score >= 60 THEN
->     SET grade = 'B';
->   ELSEIF score >= 40 THEN
->     SET grade = 'C';
->   ELSE
->     SET grade = 'F';
->   END IF;
->
->   RETURN grade;
-> END //
```

Query OK, 0 rows affected (0.00 sec)

```
mysql>
mysql> DELIMITER ;
mysql> SELECT student_name, marks, GetGrade(marks) AS grade
-> FROM Students;
```

student_name	marks	grade
Alice	85	A
Bob	72	B
Charlie	56	C
David	40	C
Eva	90	A

```
MySQL 8.0 Command Line Client
-> text VARCHAR(100)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> INSERT INTO Messages (text) VALUES
-> ('Hello World'),
-> ('MySQL String Functions'),
-> ('ChatGPT is awesome'),
-> ('Data Science');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> SELECT
-> text,
-> REPLACE(text, ' ', '_') AS replaced_text,
-> REPEAT('Ha! ', 2) AS repeated_text,
-> REVERSE(text) AS reversed_text,
-> RIGHT(text, 5) AS right_five_chars,
-> LEFT(text, 5) AS left_five_chars,
-> RPAD(text, 20, '*') AS right_padded,
-> LPAD(text, 20, '*') AS left_padded
-> FROM Messages;
+-----+-----+-----+-----+-----+-----+-----+
| text | replaced_text | repeated_text | reversed_text | right_five_chars | left_five_chars | right_padded | left_padded |
+-----+-----+-----+-----+-----+-----+-----+
| Hello World | Hello_World | Ha! Ha! | dlroW olleH | World | Hello | Hello World***** | *****Hello Worl |
| MySQL String Functions | MySQL_String_Functions | Ha! Ha! | snoitcnuF gnirtS LQSyM | tions | MySQL | MySQL String Functio | MySQL String Functi |
| ChatGPT is awesome | ChatGPT_is_awesome | Ha! Ha! | emosewa si TPGtahC | esome | ChatG | ChatGPT is awesome** | **ChatGPT is awesom |
| Data Science | Data_Science | Ha! Ha! | ecneicS ataD | ience | Data | Data Science***** | *****Data Scienc |
+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line Client
->      IF done THEN
->          LEAVE read_loop;
->      END IF;
->
->      -- Here you can process each row
->      -- For demonstration, let's just select the student name
->      SELECT CONCAT('Processing student: ', sname) AS info;
->  END LOOP;
->
->  CLOSE cur_students;
-> END //
```

Query OK, 0 rows affected (0.03 sec)

```
mysql>
mysql> DELIMITER ;
mysql> CALL ProcessStudents();
+-----+
| info |
+-----+
| Processing student: Alice |
+-----+
1 row in set (0.01 sec)

+-----+
| info |
+-----+
| Processing student: Bob |
+-----+
1 row in set (0.07 sec)

+-----+
| info |
+-----+
| Processing student: Charlie |
+-----+
1 row in set (0.10 sec)

Query OK, 0 rows affected (0.11 sec)

mysql> _
```

```
MySQL 8.0 Command Line Client
mysql> INSERT INTO AdmissionForm (student_name, date_of_birth, course_applied, admission_date)
-> VALUES
-> ('John Doe', '2003-05-15', 'Computer Science', '2025-09-01'),
-> ('Jane Smith', '2004-08-20', 'Mechanical Engineering', '2025-09-01'),
-> ('Alice Johnson', '2003-12-10', 'Civil Engineering', '2025-09-02');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM AdmissionForm;
+-----+-----+-----+-----+-----+-----+
| admission_id | student_name | date_of_birth | course_applied | admission_date | status |
+-----+-----+-----+-----+-----+-----+
| 1 | John Doe | 2003-05-15 | Computer Science | 2025-09-01 | Pending |
| 2 | Jane Smith | 2004-08-20 | Mechanical Engineering | 2025-09-01 | Pending |
| 3 | Alice Johnson | 2003-12-10 | Civil Engineering | 2025-09-02 | Pending |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT student_name, admission_date, status
-> FROM AdmissionForm
-> WHERE course_applied = 'Computer Science';
+-----+-----+-----+
| student_name | admission_date | status |
+-----+-----+-----+
| John Doe | 2025-09-01 | Pending |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> UPDATE AdmissionForm
-> SET status = 'Approved'
-> WHERE admission_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> DELETE FROM AdmissionForm
-> WHERE admission_id = 3;
Query OK, 1 row affected (0.01 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
MySQL 8.0 Command Line Cli  x  +  v

+-----+-----+-----+-----+
| bus_number | destination | departure_time | seats |
+-----+-----+-----+-----+
| BUS102     | Chennai    | 2025-09-02 10:30:00 | 50 |
+-----+-----+-----+-----+
1 row in set (0.02 sec)

mysql> SELECT U.name, B.bus_number, B.source, B.destination,
->         T.seat_no, P.payment_status, Q.qr_data
-> FROM Tickets T
-> JOIN Users U ON T.user_id = U.user_id
-> JOIN Buses B ON T.bus_id = B.bus_id
-> JOIN Payments P ON T.ticket_id = P.ticket_id
-> JOIN QR_Codes Q ON T.ticket_id = Q.ticket_id
-> WHERE T.ticket_id = 1;

+-----+-----+-----+-----+-----+-----+-----+
| name      | bus_number | source   | destination | seat_no | payment_status | qr_data |
+-----+-----+-----+-----+-----+-----+-----+
| Amit Sharma | BUS101    | Hyderabad | Bangalore   | 12      | Completed      | QR:BUS101-Seat12-User1-Ticket1 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> UPDATE Tickets
-> SET status='Cancelled'
-> WHERE ticket_id=1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT seats - COUNT(T.ticket_id) AS available_seats
-> FROM Buses B
-> LEFT JOIN Tickets T ON B.bus_id = T.bus_id AND T.status='Booked'
-> WHERE B.bus_id = 1
-> GROUP BY B.bus_id;

+-----+
| available_seats |
+-----+
| 40 |
+-----+
```

```
MySQL 8.0 Command Line Client
-> FROM Students
-> WHERE marks > (
->     SELECT AVG(marks) FROM Students
-> );
+-----+
| student_name | marks |
+-----+
| Alice        | 85    |
| Charlie      | 92    |
| Frank        | 88    |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT C.course_name, A.avg_marks
-> FROM (
->     SELECT course_id, AVG(marks) AS avg_marks
->     FROM Students
->     GROUP BY course_id
-> ) AS A
-> JOIN Courses C ON A.course_id = C.course_id;
+-----+
| course_name | avg_marks |
+-----+
| Computer Science | 75.6667 |
| Mechanical Engineering | 83.0000 |
| Civil Engineering | 92.0000 |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT student_name, marks, course_id
-> FROM Students S1
-> WHERE marks > (
->     SELECT AVG(marks)
->     FROM Students S2
->     WHERE S2.course_id = S1.course_id
-> );
```

```

--> Email VARCHAR(100) UNIQUE, -- UNIQUE
--> DeptID INT, -- FOREIGN KEY
--> FOREIGN KEY (DeptID) REFERENCES Department(DeptID) -- FOREIGN KEY
--> );
ROR 1050 (42S01): Table 'students' already exists
sql>
sql> -- 4. Insert sample data into Department
sql> INSERT INTO Department (DeptID, DeptName) VALUES
--> (101, 'Computer Science'),
--> (102, 'Electronics'),
--> (103, 'Mechanical');
ROR 1062 (23000): Duplicate entry '101' for key 'department.PRIMARY'
sql>
sql> -- 5. Insert sample data into Students
sql> INSERT INTO Students (RollNo, Name, Gender, DOB, Email, DeptID) VALUES
--> (1, 'Rahul', 'M', '2002-05-12', 'rahul@example.com', 101),
--> (2, 'Anjali', 'F', DEFAULT, 'anjali@example.com', 102), -- uses DEFAULT DOB
--> (3, 'Vikram', 'M', '2003-07-01', 'vikram@example.com', 101);
ROR 1054 (42S22): Unknown column 'DeptID' in 'field list'
sql>
sql> -- 6. Show Students table data
sql> SELECT * FROM Students;
Empty set (0.00 sec)

sql>
sql> -- 7. Show Department table data
sql> SELECT * FROM Department;
+-----+-----+
DeptID | DeptName |
+-----+-----+
101 | Computer Science |
102 | Electronics |
103 | Mechanical |
+-----+-----+

```

```
mysql> CREATE TABLE Students (  
-> RollNo INT PRIMARY KEY,  
-> Name VARCHAR(50) NOT NULL,  
-> Gender CHAR(1) CHECK (Gender IN ('M', 'F')),  
-> DOB DATE,  
-> Email VARCHAR(100) UNIQUE  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql>  
mysql> -- Insert sample data (inputs)  
mysql> INSERT INTO Students (RollNo, Name, Gender, DOB, Email)  
-> VALUES  
-> (1, 'Rahul', 'M', '2002-05-12', 'rahul@example.com'),  
-> (2, 'Anjali', 'F', '2001-11-20', 'anjali@example.com'),  
-> (3, 'Vikram', 'M', '2003-07-01', 'vikram@example.com');  
Query OK, 3 rows affected (0.02 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql>  
mysql> -- Check data (output)  
mysql> SELECT * FROM Students;  
+-----+-----+-----+-----+-----+  
| RollNo | Name  | Gender | DOB      | Email                |  
+-----+-----+-----+-----+-----+  
| 1      | Rahul | M      | 2002-05-12 | rahul@example.com    |  
| 2      | Anjali | F      | 2001-11-20 | anjali@example.com   |  
| 3      | Vikram | M      | 2003-07-01 | vikram@example.com   |  
+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)  
  
mysql>  
mysql> -- 3. ALTER TABLE: Add a new column  
mysql> ALTER TABLE Students ADD Phone VARCHAR(15);  
Query OK, 0 rows affected (0.03 sec)
```

```

MySQL 8.0 Command Line Client
mysql> -- Update Salary of Employee 102
mysql> UPDATE Employees
-> SET Salary = 54000.00
-> WHERE EmployeeID = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql> -- Oops! We want to undo the second update
mysql> ROLLBACK TO afterFirstUpdate;
ERROR 1305 (42000): SAVEPOINT afterFirstUpdate does not exist
mysql>
mysql> -- Commit changes (only first update persists)
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> BEGIN TRANSACTION;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'TRANSACTION' at line 1
mysql>
mysql> -- Delete Employee 103
mysql> DELETE FROM Employees
-> WHERE EmployeeID = 103;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> -- Something went wrong, cancel the transaction
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> -- Employee 103 is still in the table
mysql> SELECT * FROM Employees;
+-----+-----+-----+-----+-----+
| EmployeeID | FirstName | LastName | Department | Salary |
+-----+-----+-----+-----+-----+
| 101 | John | Doe | Finance | 47000.00 |
| 102 | Jane | Smith | Marketing | 54000.00 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

```
MySQL 8.0 Command Line Client
-> admission_date DATE
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> -- Create user with password
mysql> CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT SELECT, INSERT ON CollegeDB.Students TO 'user1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> REVOKE INSERT ON CollegeDB.Students FROM 'user1'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> ^N-- This will work:
-> SELECT * FROM CollegeDB.Students;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'N
SELECT * FROM CollegeDB.Students' at line 1
mysql>
mysql> -- This will fail (if INSERT was revoked):
mysql> INSERT INTO CollegeDB.Students (student_name, course, admission_date)
-> VALUES ('Alice', 'Computer Science', '2025-09-01');
Query OK, 1 row affected (0.01 sec)

mysql> -- Drop user if needed
mysql> DROP USER 'user1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE DATABASE CollegeDB;
ERROR 1007 (HY000): Can't create database 'collegedb'; database exists
mysql> CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.03 sec)

mysql> -- As user1:
mysql> SELECT * FROM CollegeDB.Students;
+-----+
| student_id | student_name | course       | admission_date |
+-----+
| 1 | Alice      | Computer Science | 2025-09-01    |
+-----+
```

```
Select MySQL 8.0 Command Line Client
mysql> SELECT * FROM HighScorers;
+-----+-----+-----+-----+
| student_id | student_name | course           | marks |
+-----+-----+-----+-----+
| 101 | Alice | Computer Science | 85 |
| 103 | Charlie | Civil Engineering | 92 |
| 105 | Eve | Electrical Engineering | 88 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE INDEX idx_marks ON Students(marks);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Students
-> WHERE marks > 80;
+-----+-----+-----+-----+
| student_id | student_name | course           | marks |
+-----+-----+-----+-----+
| 101 | Alice | Computer Science | 85 |
| 105 | Eve | Electrical Engineering | 88 |
| 103 | Charlie | Civil Engineering | 92 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SHOW INDEXES FROM Students;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | In
dex_comment | Visible | Expression |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| students | 0 | PRIMARY | 1 | student_id | A | 5 | NULL | NULL | | BTREE | | |
| students | 1 | idx_marks | 1 | marks | A | 5 | NULL | NULL | YES | BTREE | | |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.02 sec)

mysql>
```

```
MySQL 8.0 Command Line Client
| database() |
+-----+
| command |
+-----+
1 row in set (0.00 sec)

mysql> CREATE TABLE Employees (
-> EmployeeID INT PRIMARY KEY,
-> FirstName VARCHAR(50),
-> LastName VARCHAR(50),
-> Department VARCHAR(50),
-> Salary DECIMAL(10, 2)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Employees (EmployeeID, FirstName, LastName, Department, Salary)
-> VALUES
-> (101, 'John', 'Doe', 'Finance', 45000.00),
-> (102, 'Jane', 'Smith', 'Marketing', 52000.00),
-> (103, 'Alice', 'Johnson', 'HR', 48000.00);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> UPDATE Employees
-> SET Salary = 50000.00
-> WHERE EmployeeID = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> DELETE FROM Employees
-> WHERE EmployeeID = 103;
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM Employees;
+-----+-----+-----+-----+-----+
| EmployeeID | FirstName | LastName | Department | Salary |
+-----+-----+-----+-----+-----+
| 101 | John | Doe | Finance | 50000.00 |
| 102 | Jane | Smith | Marketing | 52000.00 |
+-----+-----+-----+-----+-----+
```

```
MySQL 8.0 Command Line Client
-> ('Bob Smith', 'Mechanical Engineering', '2025-09-02'),
-> ('Charlie Brown', 'Civil Engineering', '2025-09-03'),
-> ('David Wilson', 'Computer Science', '2025-09-04'),
-> ('Eve Adams', 'Electrical Engineering', '2025-09-05');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Students
-> WHERE course = 'Computer Science';
+-----+-----+-----+-----+
| student_id | student_name | course | admission_date |
+-----+-----+-----+-----+
| 1 | Alice Johnson | Computer Science | 2025-09-01 |
| 4 | David Wilson | Computer Science | 2025-09-04 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Students
-> WHERE student_name LIKE 'AS';
+-----+-----+-----+-----+
| student_id | student_name | course | admission_date |
+-----+-----+-----+-----+
| 1 | Alice Johnson | Computer Science | 2025-09-01 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM Students
-> WHERE student_name LIKE 'Nson';
+-----+-----+-----+-----+
| student_id | student_name | course | admission_date |
+-----+-----+-----+-----+
| 1 | Alice Johnson | Computer Science | 2025-09-01 |
| 4 | David Wilson | Computer Science | 2025-09-04 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Students
-> WHERE student_name LIKE 'NLSN';
+-----+-----+-----+-----+
| student_id | student_name | course | admission_date |
+-----+-----+-----+-----+
| 1 | Alice Johnson | Computer Science | 2025-09-01 |
| 3 | Charlie Brown | Civil Engineering | 2025-09-03 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Students
-> WHERE student_name LIKE 'oN';
+-----+-----+-----+-----+
| student_id | student_name | course | admission_date |
+-----+-----+-----+-----+
| 2 | Bob Smith | Mechanical Engineering | 2025-09-02 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM Students
-> WHERE course = 'Computer Science' AND student_name LIKE 'DN';
+-----+-----+-----+-----+
| student_id | student_name | course | admission_date |
+-----+-----+-----+-----+
| 4 | David Wilson | Computer Science | 2025-09-04 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line Client
-> ('Charlie', 'Electrical Engineering', 2025);
ERROR 1054 (42S22): Unknown column 'admission_year' in 'field list'
mysql>
mysql> -- Step 4: View the Records
mysql> SELECT * FROM Students;
+-----+-----+-----+-----+
| student_id | student_name | course | admission_date |
+-----+-----+-----+-----+
| 1 | Alice | Computer Science | 2025-09-01 |
| 2 | Alice | Computer Science | 2025-09-01 |
| 3 | Bob | Mechanical Engineering | 2025-09-02 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> -- Step 5: Set AUTO_INCREMENT to start from 100 for future inserts
mysql> ALTER TABLE Students AUTO_INCREMENT = 100;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql>
mysql> -- Step 6: Insert More Data (ID should now be 100+)
mysql> INSERT INTO Students (student_name, course, admission_year) VALUES
-> ('David', 'Civil Engineering', 2025),
-> ('Eve', 'Computer Science', 2024);
ERROR 1054 (42S22): Unknown column 'admission_year' in 'field list'
mysql>
mysql> -- Step 7: View All Data Again
mysql> SELECT * FROM Students;
+-----+-----+-----+-----+
| student_id | student_name | course | admission_date |
+-----+-----+-----+-----+
| 1 | Alice | Computer Science | 2025-09-01 |
| 2 | Alice | Computer Science | 2025-09-01 |
| 3 | Bob | Mechanical Engineering | 2025-09-02 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> TRUNCATE TABLE Students;
Query OK, 0 rows affected (0.06 sec)
```

```
MySQL 8.0 Command Line Client

+-----+
| Alice | Computer Science |
| Bob   | Mechanical Engineering |
| Charlie | Civil Engineering |
| David | Computer Science |
+-----+
4 rows in set (0.00 sec)

mysql> SELECT S.student_name, C.course_name
-> FROM Students S
-> INNER JOIN Courses C ON S.course_id = C.course_id;
+-----+
| student_name | course_name |
+-----+
| Alice        | Computer Science |
| Bob          | Mechanical Engineering |
| Charlie      | Civil Engineering |
| David        | Computer Science |
+-----+
4 rows in set (0.00 sec)

mysql> SELECT S.student_name, C.course_name
-> FROM Students S
-> LEFT JOIN Courses C ON S.course_id = C.course_id;
+-----+
| student_name | course_name |
+-----+
| Alice        | Computer Science |
| Bob          | Mechanical Engineering |
| Charlie      | Civil Engineering |
| David        | Computer Science |
| Eve          | NULL |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT S.student_name, C.course_name
-> FROM Students S
-> RIGHT JOIN Courses C ON S.course_id = C.course_id;
+-----+
| student_name | course_name |
+-----+
| David        | Computer Science |
| Alice        | Computer Science |
| Bob          | Mechanical Engineering |
+-----+
```

```
Select MySQL 8.0 Command Line Client
-> WHERE marks BETWEEN 70 AND 90;
+-----+-----+-----+-----+-----+
| student_id | student_name | course | marks | admission_year |
+-----+-----+-----+-----+-----+
| 1 | Alice | Computer Science | 85 | 2023 |
| 2 | Bob | Mechanical Engineering | 78 | 2024 |
| 5 | Eve | Electrical Engineering | 88 | 2025 |
| 6 | Frank | Computer Science | 75 | 2025 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Students
-> WHERE course IN ('Computer Science', 'Electrical Engineering');
+-----+-----+-----+-----+-----+
| student_id | student_name | course | marks | admission_year |
+-----+-----+-----+-----+-----+
| 1 | Alice | Computer Science | 85 | 2023 |
| 4 | David | Computer Science | 67 | 2023 |
| 5 | Eve | Electrical Engineering | 88 | 2025 |
| 6 | Frank | Computer Science | 75 | 2025 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT COUNT(*) AS total_students FROM Students;
+-----+
| total_students |
+-----+
| 6 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT AVG(marks) AS average_marks FROM Students;
+-----+
| average_marks |
+-----+
| 80.8333 |
+-----+
```

```
MySQL 8.0 Command Line Client
-> ('Bob', 'Mechanical Engineering', 78),
-> ('Charlie', 'Civil Engineering', 92);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Students;
+-----+-----+-----+-----+
| student_id | student_name | course | marks |
+-----+-----+-----+-----+
| 1 | Alice | Computer Science | 85 |
| 2 | Bob | Mechanical Engineering | 78 |
| 3 | Charlie | Civil Engineering | 92 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT student_name, marks FROM Students;
+-----+-----+
| student_name | marks |
+-----+-----+
| Alice | 85 |
| Bob | 78 |
| Charlie | 92 |
+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Students
-> WHERE marks > 80;
+-----+-----+-----+-----+
| student_id | student_name | course | marks |
+-----+-----+-----+-----+
| 1 | Alice | Computer Science | 85 |
| 3 | Charlie | Civil Engineering | 92 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```

1 row in set (0.01 sec)

```
+-----+
| CONCAT('Counter = ', counter) |
+-----+
| Counter = 2                    |
+-----+
```

1 row in set (0.03 sec)

```
+-----+
| CONCAT('Counter = ', counter) |
+-----+
| Counter = 3                    |
+-----+
```

1 row in set (0.03 sec)

```
+-----+
| CONCAT('Counter = ', counter) |
+-----+
| Counter = 4                    |
+-----+
```

1 row in set (0.03 sec)

```
+-----+
| CONCAT('Counter = ', counter) |
+-----+
| Counter = 5                    |
+-----+
```

1 row in set (0.04 sec)

Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER \$\$

mysql>

mysql> CREATE PROCEDURE WhileExample()

-> BEGIN

-> DECLARE counter INT DEFAULT 1;

->

-> WHILE counter <= 5 DO

-> -- Print counter value

-> SELECT CONCAT('Counter = ', counter);

->

-> SET counter = counter + 1;

-> END WHILE;

-> END\$\$

Query OK, 0 rows affected (0.02 sec)

mysql>

mysql> DELIMITER ;

mysql> CALL WhileExample();

```
+-----+
| CONCAT('Counter = ', counter) |
+-----+
| Counter = 1                    |
+-----+
```

1 row in set (0.00 sec)

```
+-----+
| CONCAT('Counter = ', counter) |
+-----+
| Counter = 2                    |
+-----+
```

1 row in set (0.01 sec)

```
+-----+
| CONCAT('Counter = ', counter) |
+-----+
```

```
mysql>
mysql> DELIMITER ;
mysql> CALL printNumbers(5);
```

```
+-----+
```

```
| Number |
```

```
+-----+
```

```
|      1 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
+-----+
```

```
| Number |
```

```
+-----+
```

```
|      2 |
```

```
+-----+
```

```
1 row in set (0.01 sec)
```

```
+-----+
```

```
| Number |
```

```
+-----+
```

```
|      3 |
```

```
+-----+
```

```
1 row in set (0.02 sec)
```

```
+-----+
```

```
| Number |
```

```
+-----+
```

```
|      4 |
```

```
+-----+
```

```
1 row in set (0.02 sec)
```

```
+-----+
```

```
| Number |
```

```
+-----+
```

```
|      5 |
```

```
+-----+
```

```
1 row in set (0.02 sec)
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> create table student(rollno int, name varchar(40), school varchar(30));
Query OK, 0 rows affected (0.05 sec)

mysql> insert into student(rollno, name, school)values(11223, 'saniya', 'nagarjuna'), (234, 'snehitha', 'nagarjuna'), (345, 'ayasha',
'nagarjuna');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> select from * student;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right
t syntax to use near 'from * student' at line 1
mysql> select * from student;
```

```
+-----+-----+-----+
| rollno | name   | school |
+-----+-----+-----+
| 11223 | saniya | nagarjuna |
| 234   | snehitha | nagarjuna |
| 345   | ayasha  | nagarjuna |
+-----+-----+-----+
3 rows in set (0.00 sec)
```